SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I _SR Kaarthika_ (192412294) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :SR Kaarthika

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 – Smart Medical Diagnosis System
(a) Propositional Logic Representation
Symbols

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Flu	Patient has Flu
Mea	Patient has Measles
Pne	Patient has Pneumonia

Rules
Rule 1

$$(F \wedge C) \rightarrow Flu$$

Rule 2

$$(F \wedge R) \rightarrow Mea$$

Rule 3

$$(C \wedge B) \rightarrow Pne$$

Facts

$$F$$
$$C$$
$$B$$

(b) Forward Chaining
Initial Facts
Fever
Cough
Breathlessness

Step 1
Check Rule 1
Fever ∧ Cough → Flu
Both are True.
Fire Rule 1.
Derived:
Possible Flu

Step 2
Check Rule 2
Fever ∧ Rash → Measles
Rash is False (not given).
Rule cannot fire.

Step 3
Check Rule 3
Cough ∧ Breathlessness → Pneumonia
Both True.
Fire Rule 3.
Derived
Possible Pneumonia

Final Diagnoses
Patient A has Flu.

Patient A has Pneumonia.

(c) Backward Chaining
Goal
Pneumonia
Look for rule producing Pneumonia.
Cough ∧ Breathlessness → Pneumonia
Sub-goals become
Need Cough
Need Breathlessness
Facts
Cough ✓

Breathlessness ✓
Therefore
Pneumonia proved.
Goal Reduction Tree

```
       Pneumonia
           |
   ----------
   |          |
 Cough     Breathlessness
   |          |
  TRUE        TRUE
```

Hence Goal Achieved.

CASE STUDY 2 – Autonomous Traffic Management Agent
(a) Unification
Rule 1

$$Vehicle(x) \land EmergencyType(x) \rightarrow ClearPath(x)$$

Facts
Vehicle(Ambulance)

EmergencyType(Ambulance)
Step 1
Match
Vehicle(x)

Vehicle(Ambulance)
Substitution

$$\theta_1 = \{x/Ambulance\}$$

Step 2
Apply substitution
EmergencyType(Ambulance)
Matches fact.
Final MGU

$$\boxed{\theta = \{x/Ambulance\}}$$

Therefore
ClearPath(Ambulance)
is inferred.

(b) Resolution Proof
Goal
Proceed(CarA)
Negate Goal
¬Proceed(CarA)

Convert to CNF
Rule 1
¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)
Rule 2
¬ClearPath(x) ∨ GreenSignal(x)
¬ClearPath(x) ∨ ¬RedSignal(x)
Rule 3
¬Vehicle(x)
∨ ¬Behind(x,y)
∨ ¬GreenSignal(y)
∨ Proceed(x)
Facts
Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA,Ambulance)
Negated Goal
¬Proceed(CarA)

Resolution Steps
Vehicle(Ambulance)

-

Rule1
↓
ClearPath(Ambulance)
↓
Resolve with Rule2
↓

GreenSignal(Ambulance)
↓
Use
Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)
with Rule3
↓
Proceed(CarA)
Resolve
Proceed(CarA)

¬Proceed(CarA)
↓
□ (Empty Clause)
Hence proved.

(c) Analysis
Current KB only supports **one emergency vehicle**.
If there are two ambulances,
Example
Ambulance1

Ambulance2
the KB does not specify

- which vehicle has priority
- conflict resolution
- simultaneous green signals
- route coordination

Additional Rules
Priority(x) ∧ Priority(y)
→ HigherPriority(x,y)
HigherPriority(x,y)
→ GreenSignal(x)
GreenSignal(x)
→ HoldSignal(y)
DifferentRoute(x,y)
→ AllowBoth
These rules prevent signal conflicts and ensure safe routing.

CASE STUDY 3 – Agricultural Expert System
(a) Convert into CNF
P1
Original
SoilDry → IrrigationNeeded
Remove implication
¬SoilDry ∨ IrrigationNeeded
Already CNF.
P2
IrrigationNeeded ∧ CropWheat
→ ApplyDripMethod

Remove implication
¬(I∧C) ∨ A
DeMorgan
¬I ∨ ¬C ∨ A
CNF obtained.

P3
¬ApplyDripMethod ∧ CropWheat
→ CropAtRisk
Remove implication
¬(¬A∧C) ∨ R
DeMorgan
A ∨ ¬C ∨ R
CNF.

Facts
SoilDry

CropWheat

Final CNF
¬SoilDry ∨ IrrigationNeeded

¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk

SoilDry

CropWheat

(b) Resolution Refutation
Goal
ApplyDripMethod
Negate Goal
¬ApplyDripMethod

Resolution
SoilDry

-

¬SoilDry∨IrrigationNeeded
↓
IrrigationNeeded
↓
Resolve with
¬IrrigationNeeded∨¬CropWheat∨ApplyDripMethod
↓
Using
CropWheat
↓
ApplyDripMethod
↓

Resolve
ApplyDripMethod

¬ApplyDripMethod
↓
□
Goal proved.
Resolution Tree
```
SoilDry
    \
     IrrigationNeeded
          \
CropWheat----- ApplyDripMethod
                |
          ¬ApplyDripMethod
                |
                □
```

(c) Remove SoilDry
Remaining Facts
CropWheat
Cannot derive
IrrigationNeeded
Therefore
ApplyDripMethod
cannot be proved.
Without
ApplyDripMethod
Rule 3 also cannot infer
CropAtRisk
Hence
CropAtRisk does NOT logically follow.

CASE STUDY 4 – Wumpus World
(a) Belief State
Percepts
Stench(1,2)

Breeze(1,1)

Glitter(2,2)
Rules
Stench → Adjacent Wumpus

Breeze → Adjacent Pit

Glitter → Gold

¬Wumpus ∧ ¬Pit → Safe
Using Modus Ponens
From
Stench(1,2)
Infer

Wumpus is adjacent to (1,2)
Possible cells
(1,1)

(2,2)

(1,3)

From
Breeze(1,1)
Infer
Pit adjacent to (1,1)
Possible
(1,2)

(2,1)

From
Glitter(2,2)
Infer
Gold at (2,2)

Derived Knowledge

- Wumpus adjacent to (1,2)
- Pit adjacent to (1,1)
- Gold at (2,2)

(b) Forward Chaining
Initial Facts
Stench(1,2)

Breeze(1,1)

Glitter(2,2)

Apply Rule 1
↓
Possible Wumpus
(1,1)

(2,2)

(1,3)

Apply Rule 2
↓
Possible Pit
(1,2)

(2,1)

Apply Rule 3
↓

Gold(2,2)

(c) Safety Analysis
Path
(1,1)

↓

(1,2)

↓

(2,2)
Analysis

- (1,2) may contain a Pit because of Breeze at (1,1).
- (2,2) may contain the Wumpus because of Stench at (1,2).
- Gold is located at (2,2), but the agent cannot prove that (2,2) is free of danger.
  Since Rule 4 (¬Wumpus ∧ ¬Pit → Safe) cannot be app

Final Conclusion
The path **[1,1] → [1,2] → [2,2] is not guaranteed to be safe**. The agent should gather additional percepts or explore alternative safe cells before attempting to collect the gold.